Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

Programmable Logic Devices

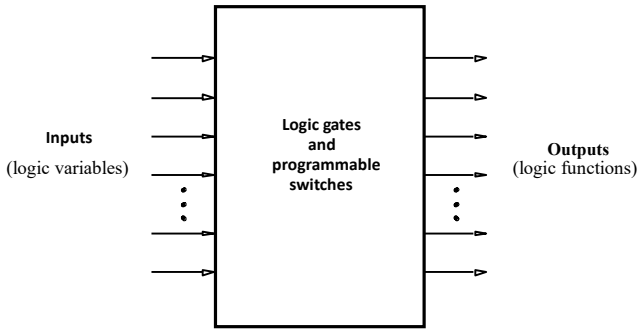
1

PLDs

- Programmable Logic Devices (PLD)
  - General purpose chip for implementing circuits
  - Can be customized using programmable switches
- Main types of PLDs
  - PLA
  - PAL
  - ROM
  - CPLD
  - FPGA
- Custom chips: standard cells, sea of gates

2

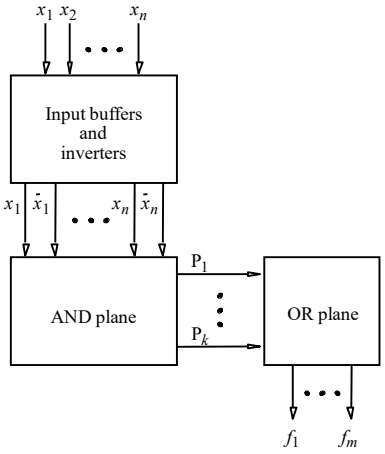
PLD as a Black Box



3

Programmable Logic Array (PLA)

- Use to implement circuits in SOP form
- The connections in the AND plane are programmable
- The connections in the OR plane are programmable



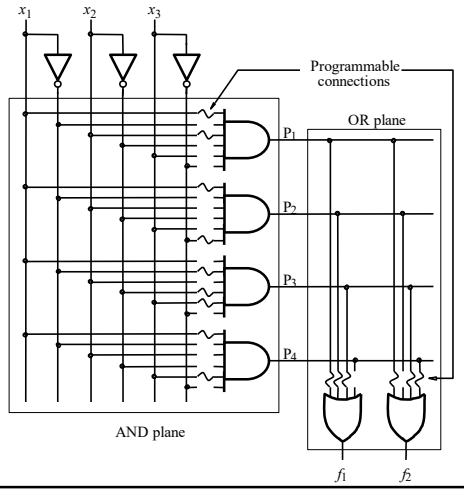
4

Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

Gate Level Version of PLA

$f_1 = x_1x_2+x_1x_3'+x_1'x_2'x_3$

$f_2 = x_1x_2+x_1'x_2'x_3+x_1x_3$



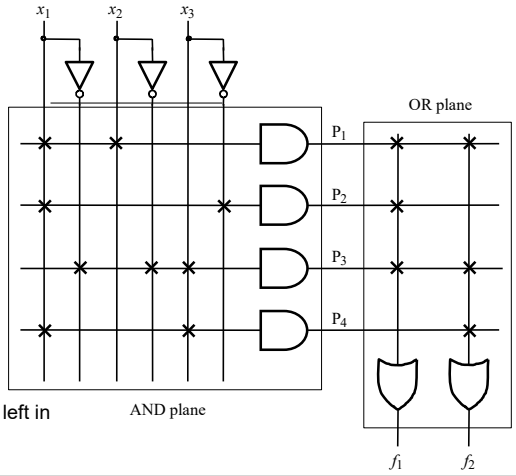
5

Customary Schematic of a PLA

$f_1 = x_1x_2+x_1x_3'+x_1'x_2'x_3$

$f_2 = x_1x_2+x_1'x_2'x_3+x_1x_3$

x marks the connections left in place after programming



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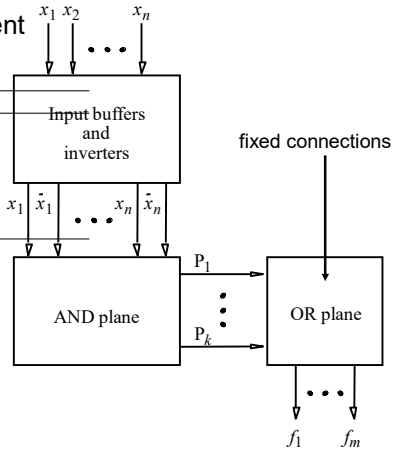
Limitations of PLAs

- PLAs come in various sizes
  - Typical size is 16 inputs, 32 product terms, 8 outputs
    - Each AND ~~gate has large fan-in~~ → this limits the number of inputs that can be ~~provided in a~~ PLA
    - 16 inputs → 3^{16} = possible input combinations; only 32 permitted (since 32 AND gates) in a typical PLA
    - 32 AND terms permitted → large fan-in for OR gates as well
      - This makes PLAs slower and slightly more expensive than some alternatives to be discussed shortly
    - 8 outputs → could have shared minterms, but not required

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Programmable Array Logic (PAL)

- Also used to implement circuits in SOP form
- The connections in the AND plane are programmable
- The connections in the OR plane are NOT programmable



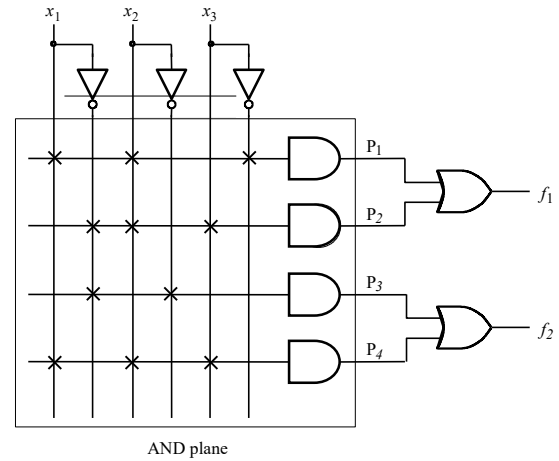
8

Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

Example Schematic of a PAL

$f_1 = x_1x_2x_3' + x_1'x_2x_3$

$f_2 = x_1'x_2' + x_1x_2x_3$



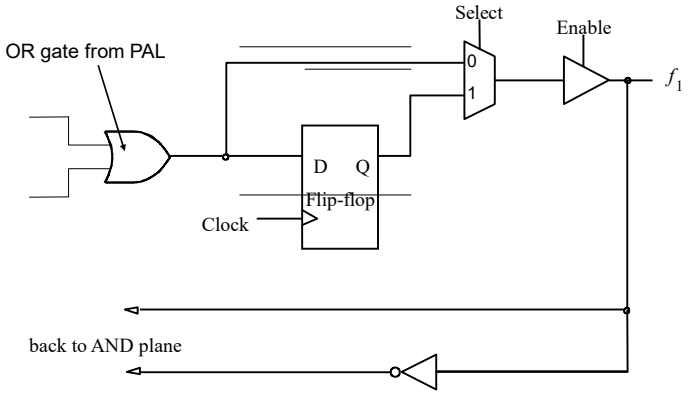
9

Comparing PALs and PLAs

- PALs have the same limitations as PLAs (small number of allowed AND terms) plus they have a fixed OR plane → less flexibility than PLAs
- PALs are simpler to manufacture, cheaper, and faster (better performance)
- PALs also often have extra circuitry connected to the output of each OR gate
  - The OR gate plus this circuitry is called a *macrocell*

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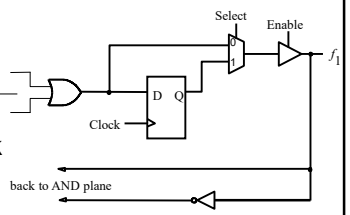
Macrocell



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Macrocell Functions

- Enable = 0 can be used to allow the output pin for f_1 to be used as an additional input pin to the PAL
- Enable = 1, Select = 0 is normal for typical PAL operation
- Enable = Select = 1 allows the PAL to synchronize the output changes with a clock pulse



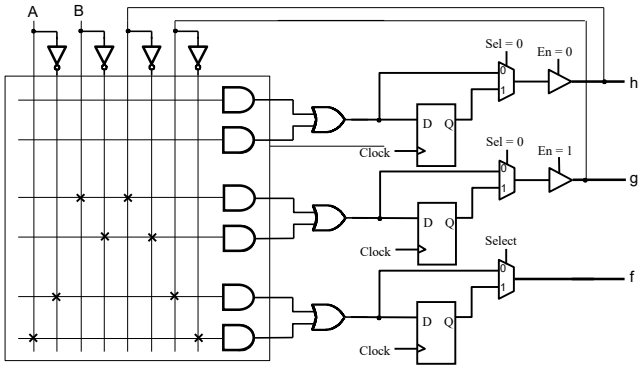
- The feedback to the AND plane provides for multi-level design

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Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

Multi-Level Design with PALs

- f = A'BC + A'B'C' + ABC' + AB'C = A'g + Ag'
  - where g = BC + B'C' and C = h below



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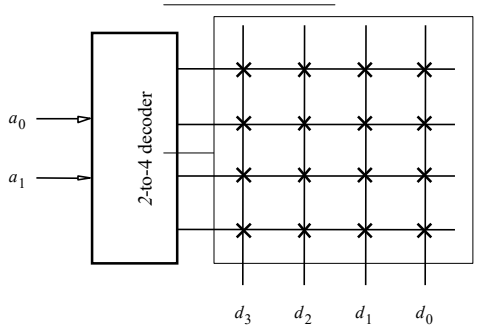
ROM

- A ROM (Read Only Memory) has a fixed AND plane and a programmable OR plane
- Size of AND plane is 2^n where n = number of input pins
  - Has an AND gate for every possible minterm so that all input combinations access a different AND gate
- OR plane dictates function mapped by the ROM

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4x4 ROM

- 2^2x4 bit ROM has 4 addresses that are decoded



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Programming SPLDs

- PLAs, PALs, and ROMs are also called SPLDs – Simple Programmable Logic Devices
- SPLDs must be programmed so that the switches are in the correct places
  - CAD tools are usually used to do this
    - A fuse map is created by the CAD tool and then that map is downloaded to the device via a special programming unit
  - There are two basic types of programming techniques
    - Removable sockets on a PCB
    - In system programming (ISP) on a PCB
      - This approach is not very common for PLAs and PALs but it is quite common for more complex PLDs

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Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

An SPLD Programming Unit

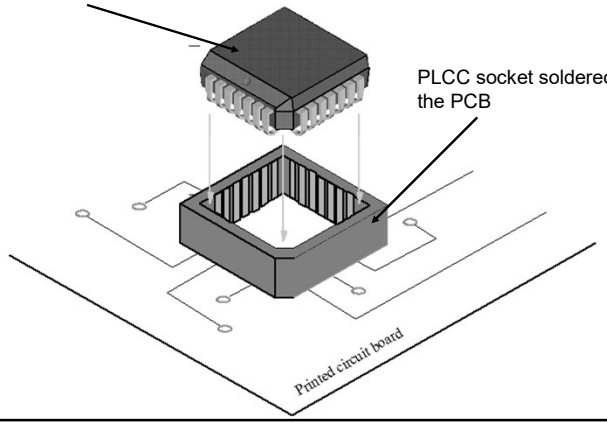


- The SPLD is removed from the PCB, placed into the unit and programmed there

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Removable SPLD Socket Package

- PLCC (plastic-leaded chip carrier)



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In System Programming (ISP)

- Used when the SPLD cannot be removed from the PCB
- A special cable and PCB connection are required to program the SPLD from an attached computer
- Very common approach to programming more complex PLDs like CPLDs, FPGAs, etc.

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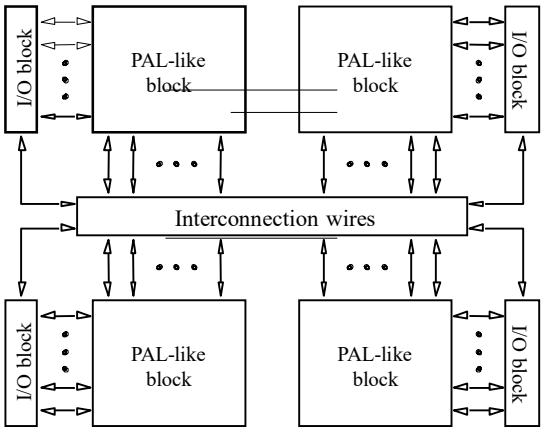
CPLD

- Complex Programmable Logic Devices (CPLD)
- SPLDs (PLA, PAL) are limited in size due to the small number of input and output pins and the limited number of product terms
  - Combined number of inputs + outputs < 32 or so
- CPLDs contain multiple circuit blocks on a single chip
  - Each block is like a PAL: PAL-like block
  - Connections are provided between PAL-like blocks via an interconnection network that is programmable
  - Each block is connected to an I/O block as well

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Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

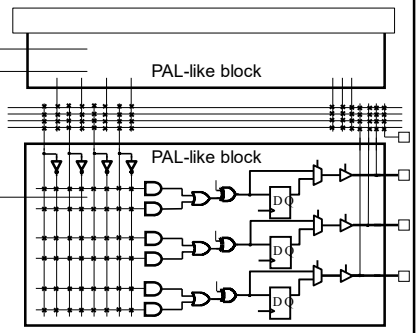
Structure of a CPLD



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Internal Structure of a PAL-like Block

- Includes macrocells
  - Usually about 16 each
- Fixed OR planes
  - OR gates have fan-in between 5-20
- XOR gates provide negation ability
  - XOR has a control input



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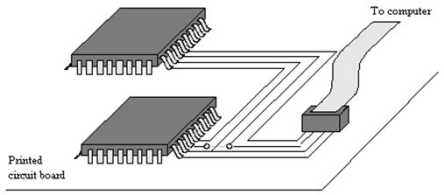
More on PAL-like Blocks

- CPLD pins are provided to control XOR, MUX, and tri-state gates
- When tri-state gate is disabled, the corresponding output pin can be used as an input pin
  - The associated PAL-like block is then useless
- The AND plane and interconnection network are programmable
- Commercial CPLDs have between 2-100 PAL-like blocks

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Programming a CPLD

- CPLDs have many pins – large ones have > 200
  - Removal of CPLD from a PCB is difficult without breaking the pins
  - Use ISP (in system programming) to program the CPLD
  - JTAG (Joint Test Action Group) port used to connect the CPLD to a computer

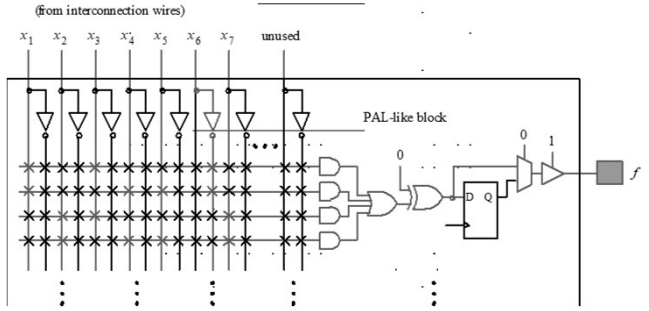


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Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

Example CPLD

- Use a CPLD to implement the function
  - $f = x_1x_3x_6' + x_1x_4x_5x_6' + x_2x_3x_7 + x_2x_4x_5x_7$



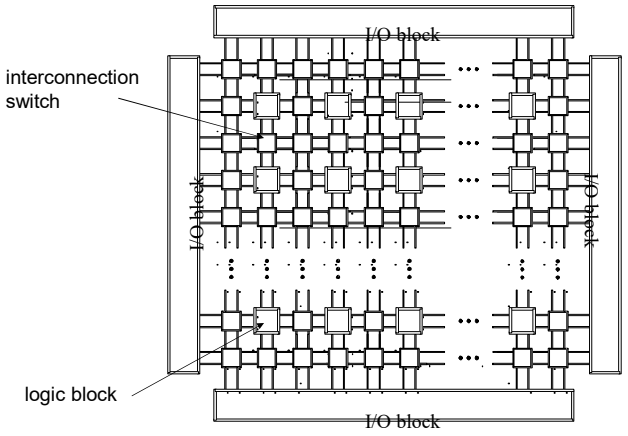
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FPGA

- SPLDs and CPLDs are relatively small and useful for simple logic devices
  - Up to about 20000 gates
- Field Programmable Gate Arrays (FPGA) can handle larger circuits
  - No AND/OR planes
  - Provide logic blocks, I/O blocks, and interconnection wires and switches
  - Logic blocks provide functionality
  - Interconnection switches allow logic blocks to be connected to each other and to the I/O pins

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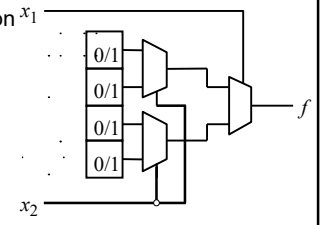
Structure of an FPGA



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LUTs

- Logic blocks are implemented using a lookup table (LUT)
  - Small number of inputs, one output
  - Contains storage cells that can be loaded with the desired values
  - A 2 input LUT uses 3 MUXes to implement any desired function of 2 variables
    - Shannon's expansion at work!



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Öğrenci kopyasıdır. İzinsiz çoğaltılamaz ve dağıtılamaz.

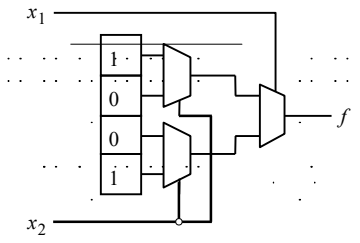
Example 2 Input LUT

x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	1

$f = x_1'x_2' + x_1x_2$, or using Shannon's expansion:

$f = x_1'(x_2') + x_1(x_2)$

$= x_1'(x_2'(1) + x_2(0)) + x_1(x_2'(0) + x_2(1))$



3 Input LUT

- 7 2x1 MUXes and 8 storage cells are required
- Commercial LUTs have 4-5 inputs, and 16-32 storage cells

